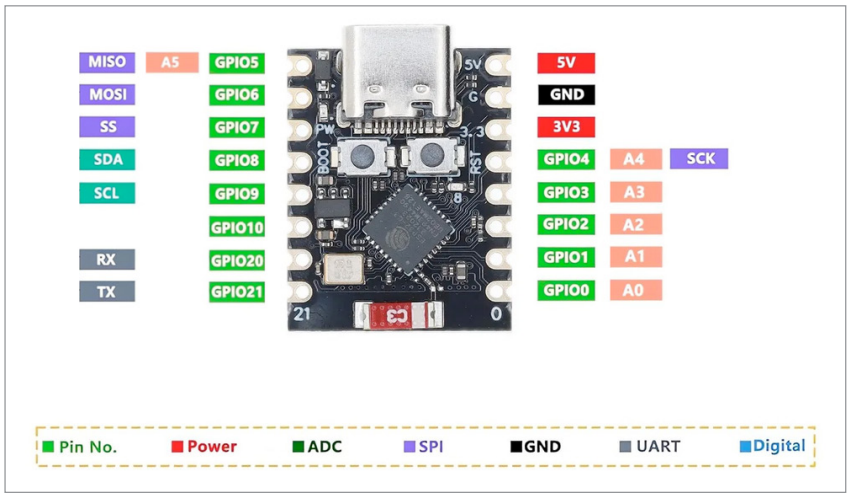


Figure 2: ESP32 C3 Super Mini pin diagram

could be done through the Thonny IDE or a command line tool called esptool. We'll use the esptool as that's more suitable for non-interactive automation pipelines. You can try flashing the board using the Thonny IDE on your own. Just plug the board with USB C data cable into your machine, select the ESP32 board detected in the right bottom corner of the IDE, click on *Configure interpreter*, select appropriate dropdown entries in the launched *Thonny options* dialog box *Interpreter* tab, click on the *Install or update MicroPython (esptool)* hyperlink, select appropriate dropdown entries in the launched *Install MicroPython (esptool)* dialog box, click on the *Install* button and you're done. Both the dialog boxes with dropdown entries are shown in Figure 3 to kill any confusions. Please note that your device port may not be ACM0 in case multiple IoT boards are plugged into your machine. Just adjust for the correct port value in that case.

Let's first prepare a Docker image to run the esptool to flash the required MicroPython firmware through the commandline. Create the *Dockerfile_Esptool* shown below and execute the command *docker build . -f Dockerfile_Esptool -t esptool* to create esptool Docker image.

```
#--- Start of Dockerfile_Esptool ---

FROM python:3.14-alpine

SHELL ["/bin/ash", "-o", "pipefail",
"-c"]
WORKDIR /tmp
RUN apk add --no-cache curl \
  && pip install esptool \
  && mkdir /etc/esp32c3sm

WORKDIR /etc/esp32c3sm
ENTRYPOINT ["esptool"]
CMD ["-h"]

#--- End of Dockerfile_Esptool ---
```

Now download the latest version of the MicroPython firmware on your machine by executing the command:

```
wget "https://micropython.org/$(curl
-s https://micropython.org/download/
LOLIN_C3_MINI/|grep '\.bin'|grep -v
preview|grep href|head -1|awk -F '"'
'{print $2}')" -o log -O /tmp/ESP32_
GENERIC_C3_Latest.bin
```

You can also get into a running esptool container to download the MicroPython firmware in case you

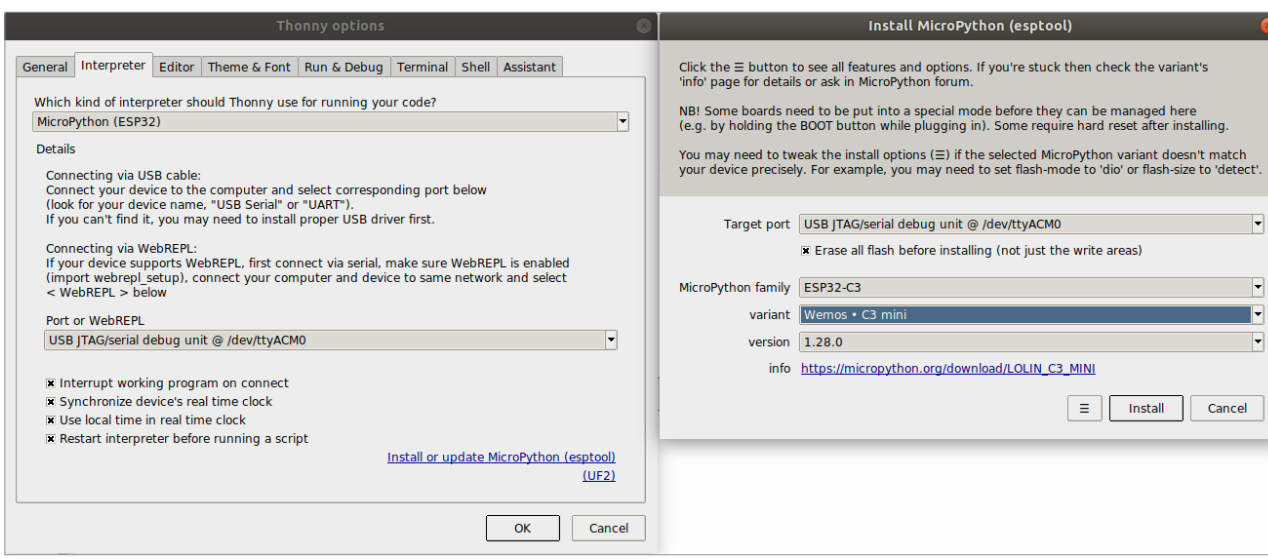


Figure 3: Flashing ESP32 C3 Super Mini through Thonny IDE